

Medical abbreviation bake-off - three local model sizes

CS496 Project 0 | MEASURED FINAL TEST

Run: 20260914T001358897252Z | Three-local-model adaptation of the API/API/local brief.

1. Task and example

Task: expand one medical abbreviation from a short UK medical-record context. The gold answer is hidden from the model and scoring is automatic.

Example: context: An AMHP coordinated the statutory mental health assessment. Question: What does AMHP mean here? Expected: approved mental health professional.

2. Data

50 test items; the project has 50 test and 10 development items with disjoint abbreviations. Labels come from the NHS abbreviation glossary; contexts are synthetic and contain no patient data. Review records: two distinct reviewer names recorded; recorded names are not independent proof of review. Development items were used only for setup; final accuracy uses the 50-item frozen test set.

3. Setup

Models: qwen2.5:1.5b, qwen2.5:3b, qwen2.5:7b. All three run locally through Ollama on the same machine.

Settings: temperature 0; output limit 64 tokens; same prompt, same question order, same parser; one sequential call per item; no retries, tools, retrieval or chat history. Scoring trims outer whitespace and ignores letter case, but wording otherwise must match the canonical answer exactly. Timeouts, truncation and malformed outputs count as wrong.

Hardware: CPU AMD Ryzen 5 5600X 6-Core Processor; GPU NVIDIA GeForce RTX 3060; RAM 15.9 GB. Full detected model metadata is saved in results/hardware.md.

Prompt: Expand the medical abbreviation in the supplied UK medical-record context. Return ONLY the full medical term on one line. Do not include the abbreviation, an explanation, quotes, JSON, a label such as "Answer:", or a final full stop. Use the standard UK medical expansion indicated by the context. Context: An AMHP coordinated the statutory mental health assessment. Question: What does AMHP mean here? Answer:

4. Results

Model	Correct	p50 / p95 ms	USD / 1k	tokens/s	Parse/Refusal/Timeout/Other
qwen2.5:1.5b	16/50 (32%)	2228 / 2413	5.0950	50.6	33/0/0/0
qwen2.5:3b	16/50 (32%)	2223 / 2361	5.0956	58.7	33/0/0/0
qwen2.5:7b	27/50 (54%)	2272 / 2387	5.0989	37.0	22/0/0/0

Parse includes out-of-vocabulary answers. Fixed refusal prefixes are counted separately; unrecognised refusal wording can remain a parse error. Latency is full wall-clock request time. p50 and p95 include every attempted request. Cost/1k uses each model's measured throughput plus the same hardware/electricity and operator-time assumptions. All 50 attempts per model remain in the accuracy denominator.

Evidence, decision and scaling

4. Results - three wrong answers per model

qwen2.5:1.5b - 34 failures

test_001 (AMHP): expected approved mental health professional; output Assessment and Management of People with Mental Health Problems [parse_error].

test_003 (BNO): expected bowels not open; output Bowel Number [parse_error].

test_004 (BO): expected bowels open; output bowel [parse_error].

qwen2.5:3b - 34 failures

test_001 (AMHP): expected approved mental health professional; output Adult Mental Health Practitioner [parse_error].

test_003 (BNO): expected bowels not open; output Bowel Not Operating [parse_error].

test_004 (BO): expected bowels open; output Bowel obstruction [parse_error].

qwen2.5:7b - 23 failures

test_003 (BNO): expected bowels not open; output Bowel Not Open [parse_error].

test_004 (BO): expected bowels open; output Bowel Opening [parse_error].

test_007 (CSU): expected catheter stream urine sample; output Catheterised Suprapubic Urine [parse_error].

5. Choice and when we would change it

No model meets the project's 95% accuracy + p95<10s target. For this benchmark, choose the highest-accuracy model: qwen2.5:7b at 27/50 (54.0%).

We would change the choice if a smaller model reached the same required accuracy with materially lower p95 latency/cost, if the larger model gained enough accuracy to justify its slower throughput, or if deployment hardware/cost assumptions changed.

6. Cost at 100x traffic and break-even

All candidates are local: no paid API requests. Estimated hardware/electricity: \$0.15/hour; operator time: 1 hour/month at \$5/hour. Baseline: 1,000 requests/month; 100x: 100,000. Costs assume hardware is charged only for active processing time.

qwen2.5:1.5b: today \$5.095; at 100x \$14.502; capacity 252577 requests/month; 100x fits one machine: True.

qwen2.5:3b: today \$5.096; at 100x \$14.562; capacity 250990 requests/month; 100x fits one machine: True.

qwen2.5:7b: today \$5.099; at 100x \$14.893; capacity 242601 requests/month; 100x fits one machine: True.

For an API price A per request, local variable cost v and monthly labour F, break-even $N = F/(A-v)$, only if $A > v$ and capacity allows. No API was measured, so a numeric API-vs-local break-even is unavailable.

Break-even note: because this local-only variant compares three self-hosted models rather than an API model against a self-hosted model, there is no API-vs-local break-even volume to compute. If the original brief is enforced literally, a separate top-API + cheap-API + local run is required to satisfy that item.

Limitations

This benchmark measures canonical abbreviation expansion, not clinical competence. Synthetic contexts and 50 test cases limit generalisation; one item changes accuracy by two percentage points. The final run is preserved exactly and is not edited after seeing outputs.

Sources: NHS, nhs.uk/nhs-app/help/understanding-abbreviations/ (source recorded in dataset); Ollama, docs.ollama.com/api/chat. Full responses and model digests remain in results/runs/.